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Linking metal-organic cages pairwise as a design approach for assembling multivariate crystalline materials

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Abstract

Using metal-organic cages (MOCs) as preformed supermolecular building-blocks (SBBs) is a powerful strategy to design functional metal-organic frameworks (MOFs) with control over the pore architecture and connectivity. However, introducing chemical complexity into the network via this route is limited as most methodologies focus on only one type of MOC as the building-block. Herein we present the pairwise linking of MOCs as a design approach to introduce defined chemical complexity into porous materials. Our methodology exploits preferential Rh-aniline coordination and stoichiometric control to rationally link Cu4L4 and Rh4L4 MOCs into chemically complex, yet extremely well-defined crystalline solids. This strategy is expected to open up significant new possibilities to design bespoke multi-functional materials with atomistic control over the location and ordering of chemical functionalities.

Keywords

Metal-organic cage, Metal-organic framework, multivariate, linking, porous

Linking metal-organic cages pairwise as a design approach for assembling multivariate crystalline materials

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ABSTRACT: Using metal-organic cages (MOCs) as preformed supermolecular building-blocks (SBBs) is a powerful strategy to design functional metal-organic frameworks (MOFs) with control over the pore architecture and connectivity. However, introducing chemical complexity into the network via this route is limited as most methodologies focus on only one type of MOC as the building-block. Herein we present the pairwise linking of MOCs as a design approach to introduce defined chemical complexity into porous materials. Our methodology exploits preferential Rh-aniline coordination and stoichiometric control to rationally link Cu_4L_4 and Rh_4L_4 MOCs into chemically complex, yet extremely well-defined crystalline solids. This strategy is expected to open up significant new possibilities to design bespoke multi-functional materials with atomistic control over the location and ordering of chemical functionalities.

INTRODUCTION

The ability to control the solid-state ordering of mixed functionalities opens up exciting opportunities to design bespoke materials with multi-functional or cooperative properties.[1-3] This is especially true in the case of metal-organic frameworks (MOFs), which have promising applications in topical areas such as gas separation and storage,[4–6] catalysis[7,8] and biomedicine.[9–11] The synthesis of MOFs with mixed-functionalities – that is, multivariate structures composed of multiple different metal ions and ligands – is largely reliant on an initial 'one-pot' synthesis of simple ligand and metal salt precursors.[1, 12-14] As the number and type of precursors increases, the crystallisation of multiple phases becomes an issue due to a lack of control over the competing assembly processes that occur *in-situ*.

In contrast, multi-functional architectures in biological systems are formed by the step-wise ordering of preorganised and well-defined subunits.[15, 16] In mimicking this approach, supermolecular building blocks (SBBs) such as molecular cages have been ordered into complex crystalline materials via hydrogen bonding,[17] chiral-recognition,[18] and ionic interactions.[19] Whilst these advances are significant, predicting the crystal structure for solids formed by non-covalent interactions is a long-standing challenge in solid-state chemistry.[20,21]

Metal-organic cages (MOCs) based on $[M_2(COO)_4(solvent)_2]$ ($M = Cu^{2+}$, Rh^{2+} and Cr^{2+}) paddlewheel nodes are a powerful class of SBBs because they offer well-defined assembly options through their exterior coordination sites and compatible ligand-based functionalities.[22,23] However, the methodologies for synthesising MOFs from preformed MOCs have been limited to only one type of MOC as the building-block.[24-26]

Zhou first demonstrated that preformed $Cu_{12}L_{12}$ octahedra can be linked with 4,4'-bipyridine to generate crystals of an interpenetrated MOF (Fig. 1a).[24] More recently, Maspoch and co-workers showed that $Rh_{24}L_{24}$ MOCs can be utilised as SBBs to generate highly-connected MOFs via coordination of metal clusters to the cage exterior (Fig. 1b).[27] The coordinative linking of two different MOCs, however, represents a significant advancement in the design of multi-functional porous solids, but is hitherto unrealized. This may be due to the increased complexity of such a system – in terms of the challenge of directing the selective coordination of one MOC to another

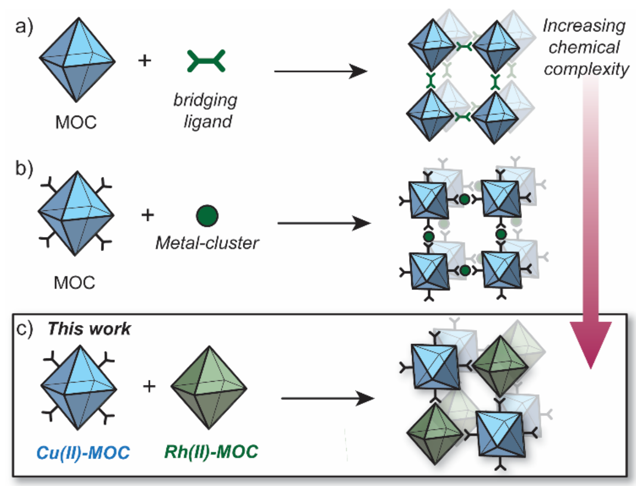

Figure 1. Strategies to synthesise MOFs from preformed MOCs; a) utilising bridging ligands to link MOCs via the available coordination sites;[24] b) linking surface-functionalised MOCs via a metal cluster;[27] c) *this work*: linking a surface-functionalised Cu^{2+} MOC to a Rh^{2+} MOC via preferential Rh-aniline coordination.

Herein we report the pairwise linking of two different MOCs as a design approach for generating an unprecedented class of multivariate crystalline materials (Fig. 1c). The MOC building-blocks utilised in this study are based on a M_4L_4 lantern architecture ($M = Cu^{2+}$ or Rh^{2+}) and their pairwise linking is achieved by preferential Rh-aniline coordination. The hallmark of this approach is the ability to preprogram the positioning of multiple metal ions and ligand-based functionalities within a structurally well-defined porous material.

RESULTS AND DISCUSSION

Synthesis of MOC building-blocks

As part of our efforts to design MOCs as SBBs, we recently reported $[Cu_4L^1_4]$ – denoted here as **1-Cu**.[28] **1-Cu** is a soluble cage structure composed of two $[Cu_2(COO)_4(solvent)_2]$ paddlewheel nodes and four ligands based on a fluoroaniline backbone (Fig. 2a). **1-Cu** is a rare example of a MOC that possesses exterior amine groups and vacant coordination sites yet remains stable in solution.[29–31]

In order to target a heterogeneous bimetallic system, we prepared a kinetically inert $[Rh_4L^2_4]$ cage as the linking partner. We selected the aldehyde-functionalised ligand L^2 based on the established chemistry of the $[Cu_4L^2_4]$ MOC analogue, as well as the ability to explore covalent linking options.[32] Thus, $[Rh_4L^2_4]$ (**2-Rh**, Figure 2a) was prepared by combining L^2, $Rh_2(OAc)_4$ and Na_2CO_3 in DMA and heating the resultant mixture at 85 °C for 32 h. Single-crystals of **2-Rh** were grown by slow-vapor diffusion of MeCN into a DMF solution of the cage. Single-crystal X-ray diffraction (SCXRD) confirmed the M_4L_4 structure and revealed that the axial paddlewheel sites of the MOC are occupied by MeCN ligands (Figure S25).

Synthesis and structure of mixed-cage MOF 3

In conceptualizing **1-Cu** and **2-Rh** as SBBs, both MOCs represent planar four-connecting nodes when considering the positioning and orientation of their covalent functionalities. However, when combining the two MOCs in DMF, we observed a crystallisation phenomenon that was dominant over their covalent reactivity. A microcrystalline precipitate with a Cu:Rh ratio of 1:2 (as determined by EDX analysis) was isolated from a DMF mixture of **1-Cu** and **2-Rh** after 2 h at room temperature. This co-crystallisation, however, did not occur in DMA as the solvent, which enabled us to grow larger crystals of the material for SCXRD analysis.

Slow-vapor diffusion of ethyl acetate into the 1:2 mixture of **1-Cu** and **2-Rh** in DMA resulted in the formation of ~200 μm-sized plate-like crystals of $\{[\mathbf{1\text{-}Cu}]\cdot[\mathbf{2\text{-}Rh}]_2\}_n$, (herein denoted as **3**) after 5 days. Synchrotron SCXRD revealed that **3** crystallizes in the tetragonal space group $I4/m$. The asymmetric unit contains 1/8 of the structure **1-Cu** and 1/4 of the structure of **2-Rh** (Figure 2b), which is in agreement with the 1:2 Cu:Rh stoichiometry observed by EDX. The X-ray structure reveals that **1-Cu** and **2-Rh** crystallise through coordinative linking; the exterior coordination sites of **2-Rh** are both occupied by an aniline donor from a separate molecule of **1-Cu** ($D_{N1\text{-}Rh1}$: 2.23 Å; C14-N1-Rh1: 109.0° - Figure 2a). Their connectivity is both highly directional and rational; **1-Cu** behaves as a four-connecting node through its exterior aniline donors, whilst **2-Rh** behaves as a two-connecting node through the two exterior paddlewheel sites (Figure 2c). This combination produces a (4,4)-net, where the 2-D layers pack in a staggered ABAB fashion (Figure 3c). The eclipsed orientation of **2-Rh** along the *c* axis creates channels approximating the dimensions of internal cavity of the MOC (~7 × 8 Å).

The bulk purity of **3** was confirmed by Powder X-ray Diffraction (PXRD). A Rietveld refinement of the data yielded R_{wp} and GoF (Goodness of Fit) parameters of 3.89 and 1.15 respectively, indicating that the bulk sample is phase pure and in excellent agreement with the SCXRD structure (Figure 3a, S10). This is remarkable given the propensity of **1-Cu** and **2-Rh** to crystallize as discrete structures under similar conditions.[28, 32] The 1:2 stoichiometry of **1-Cu** and **2-Rh** as observed by SEM/EDX analysis could also be validated by ^1H NMR spectroscopy. A sample of **3** was dissolved in DMF-d_7 by heating and the characteristic resonances from the two discrete MOCs could be compared in the ^1H NMR spectrum (Figure S5). Allowing the solution to stand at 25 °C for 1 day resulted in the re-crystallization of **3** (Figure 3b, S12).

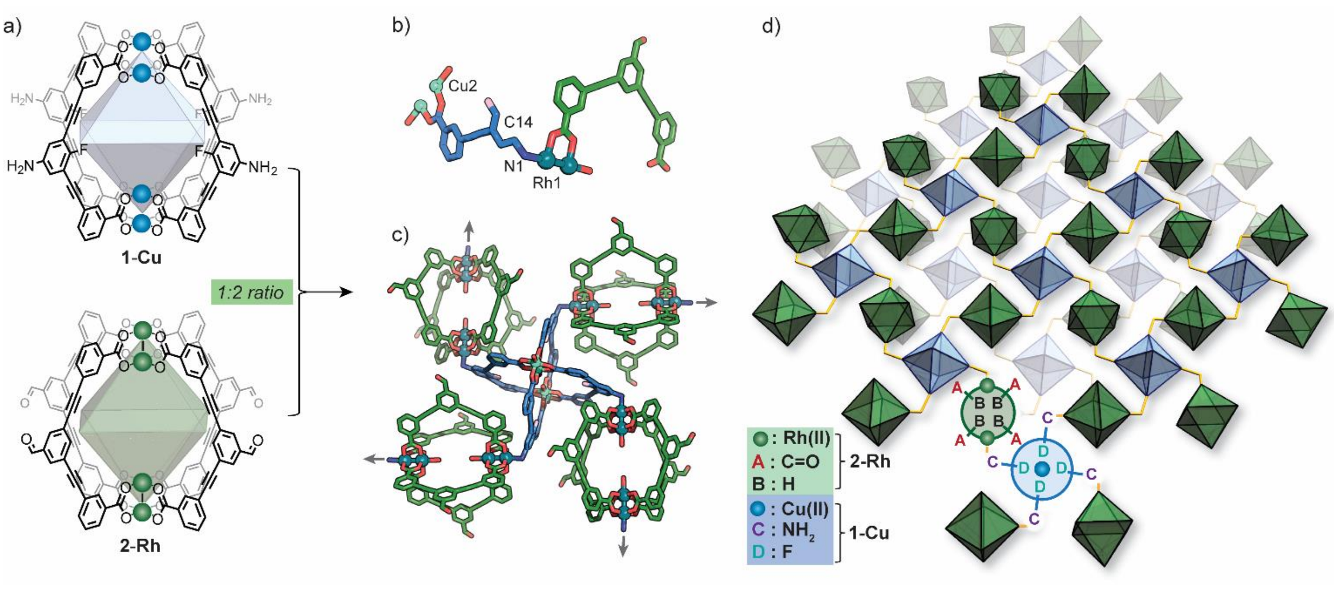

Figure 2: a) The molecular structure of **1-Cu** and **2-Rh** and their pairwise linking into a 2-D multivariate MOF $\{[\mathbf{1\text{-}Cu}]\cdot[\mathbf{2\text{-}Rh}]_2\}_n$ (**3**); a) the asymmetric unit of **3**; b) a view of **1-Cu** and **2-Rh** as four and two connecting nodes, respectively; c) a perspective view of the 2-D structure of **3**, where the two MOCs are represented as polyhedra and their ordered functionalities (Rh^{2+} and Cu^{2+}, A – D) are annotated.

Whilst multiple MOC entities may be isolated within a MOF structure through in-situ assembly,[33, 34] this is the first example where two discrete, preformed MOCs have been linked into a multivariate MOF material. Here, the location and coordination environment of the Cu^{2+} and Rh^{2+} metal ions is both predictable and extremely well-defined. In addition, the checkerboard arrangement of **1-Cu** and **2-Rh** results in the precise ordering of two alternating pore environments, and therefore, multiple ligand-based functionalities (A – D, Figure 2d).

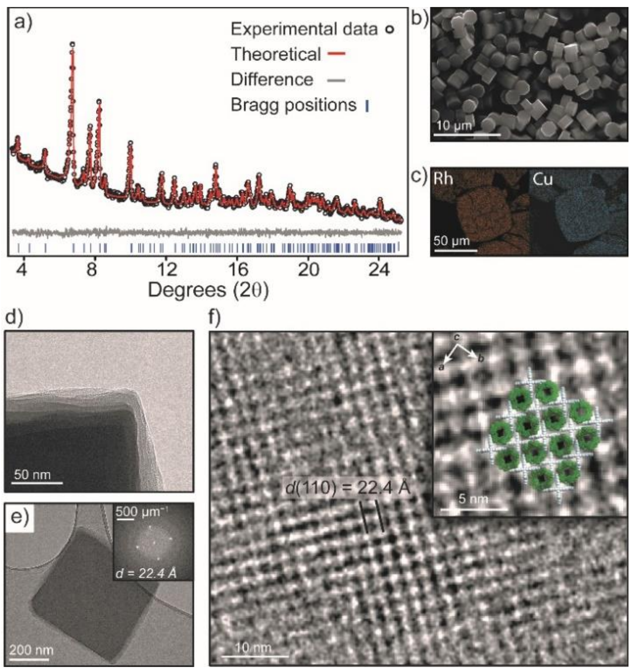

Figure 3: a) Rietveld refinement of the PXRD data of **3**; b) SEM image of **3** obtained from a DMF recrystallization; c) EDX elemental mapping of Cu and Rh within crystals of **3**; TEM images of **3** d) the edge of a crystal, showing misalignment of layers; e) the FFT pattern of a selected crystal; f) an enlarged image of the same crystal showing the superimposition of the crystal structure of **3** along the c axis.

Activation of **3** was carried out by super-critical CO_2 exchange of an acetone-solvated sample. Whilst PXRD revealed a reduction in long-range order (Figure S12), transmission electron microscopy (TEM) clearly showed that the MOC ordering within the layers of **3** is largely maintained. An image focused at the edge of a ~400 nm sized crystal clearly shows an irregular alignment of the layers in the activated sample (Figure 3d). Nevertheless, the Fast Fourier Transform (FFT) pattern in Figure 3e displays prominent peaks that correspond to a *d* spacing of 22.4 Å. This *d* spacing is in good agreement with the hkl 110 plane of **3** (hkl 110 = 22.9 Å) which represents the distance between **1-Cu** along the crystallographic *c* axis. In Figure 3f, the bright regions in the TEM image represent the interconnected **1-Cu** and **2-Rh** MOCs, where **1-Cu** is oriented vertically to the direction of the electron beam. The dark areas correspond to the pores of **2-Rh**, which is oriented perpendicular to the electron beam.

The crystal packing and pore-structure of **3** is clearly dictated by the coordinative linking of **1-Cu** and **2-Rh**. This is in stark contrast to most molecular cage solids, where weak non-covalent interactions are responsible for their solid-state ordering and resulting pore-structure.[35, 36] As such, we sought to evaluate the porosity of **3** by N_2 and CO_2 adsorption measurements at 77 K and 195 K, respectively. The N_2 isotherm of **3** displays a typical type-I adsorption profile with a Brunauer-Emmett-Teller surface area (SA_{BET}) of 422 m^2/g. The pore-size distribution (PSD) derived from the low-pressure region shows a maximum at 8.0 Å (Figure S15) which agrees well with the expected dimensions of the internal cavity of **2-Rh**. In contrast, the two discrete MOC solids are non-porous to N_2 at 77 K (Figure 5), presumably because of the random structural aggregation that accompanies their activation (both solids are amorphous – Figure S9 and ref. 28). The CO_2 isotherms measured 195 K revealed that **1-Cu**, **2-Rh**, and **3** display type I profiles (Figure S16). A marked increase in surface area is evident for **3** when comparing to the discrete MOCs; SA_{BET} of **3** = 359 m^2/g (ca. 95 and 282 m^2/g for **1-Cu** and **2-Rh**, respectively).

Stoichiometric control over the MOC network structure

Given the molecular nature of **1-Cu** and **2-Rh**, we sought to examine whether their connectivity can be dictated by their relative stoichiometry in solution. Owing to the directionality of their coordinative linking, we predicted that a change in MOC stoichiometry should result in a change in network structure. Indeed, whilst a 1:2 stoichiometry of **1-Cu** and **2-Rh** gave crystals of the 2D MOF **3**, a completely different mixed-cage solid (denoted here as **4**) was obtained from the crystallisation of a 1:1 MOC mixture.

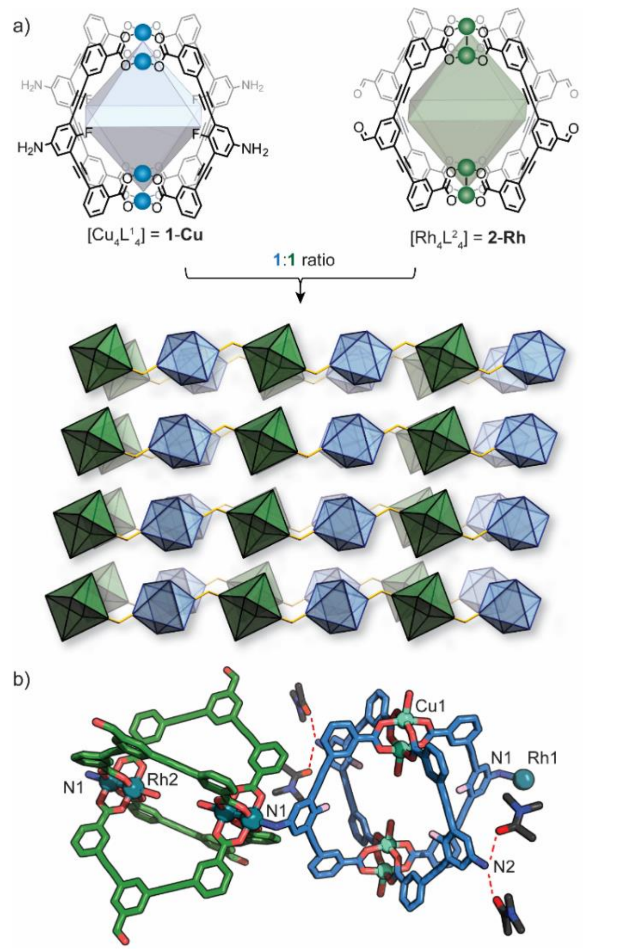

Figure 4. a) assembly of {[**1-Cu**]·[**2-Rh**]}$_n$ (**4**) from a 1:1 mixture of **1-Cu** (left) and **2-Rh** (right); b) A perspective view of

the crystal packing **4** (MOCs represented as polyhedra); c) a view of the coordination of **1-Cu** to **2-Rh**. The bulk characterisation of **4** is described in the supporting information.

Synchrotron SCXRD revealed that **4** crystallizes in the triclinic space group *P*-1, with a half of each M_4L_4 structure in the asymmetric unit. Again, both exterior axial sites of **2-Rh** are coordinated by an aniline donor from a separate molecule of **1-Cu** ($D_{N\text{-}Rh}$: 2.22 Å; C-N-Rh: 113.2° - Figure 4b). Due to the change in stoichiometry, only two of the four aniline donors of **1-Cu** participate in coordination, and the remaining two donors hydrogen bond with co-crystallized DMA solvent; $D_{NH\cdots O}$ = 2.02 Å and 2.46 Å, 151.1° and 166.2°, respectively. Thus, both MOCs act as linear two-connecting nodes to give rise to a 1-D coordination polymer with a formula of {[**1-Cu**]·[**2-Rh**]}$_n$ (Fig 4a).

Elucidating the preference for Rh-aniline coordination

To rationalize the observed preference for aniline-Rh^{2+} coordination, we carried out DFT calculations on representative paddlewheel complexes (full details in Supporting Information). The calculated energy for the ligand exchange reaction shown in Figure 5a amounts to ΔE = −5.4 kcal mol−1 which supports a thermodynamic argument in favour of preferential coordination of aniline by the Rh^{2+} paddlewheel complex (as compared to the Cu^{2+} counterpart). For the optimized complexes, the Cu-N bond is ~0.7 Å shorter compared to the Rh-N bond, presumably due to the smaller ionic radius of Cu^{2+} and the trans effect of the Rh^{2+} complex.[37] This agrees well with the crystallographic results where the Rh-N aniline bond of the linked MOCs is longer by ~0.4 Å compared to Cu-N.[28] Furthermore, the optimized geometry of $[Cu_2(PhCOO)_4(PhNH_2)_2]$ exhibits Cu-O-O-Cu dihedral angles between 14.7° and 15.9° while the corresponding the Rh-O-O-Rh dihedral angles are below 1.3° (Fig. 5b and S29). The shorter Cu-N bond length and distortion of $[Cu_2(PhCOO)_4(PhNH_2)_2]$ compared to the Rh analogue therefore suggests that the observed preference is associated with a reduced steric hindrance for Rh-aniline coordination. It is noteworthy that $Rh_{24}L_{24}$ MOCs have recently been used to separate regioisomers of methyl pyridine based on their steric hindrance.[38]

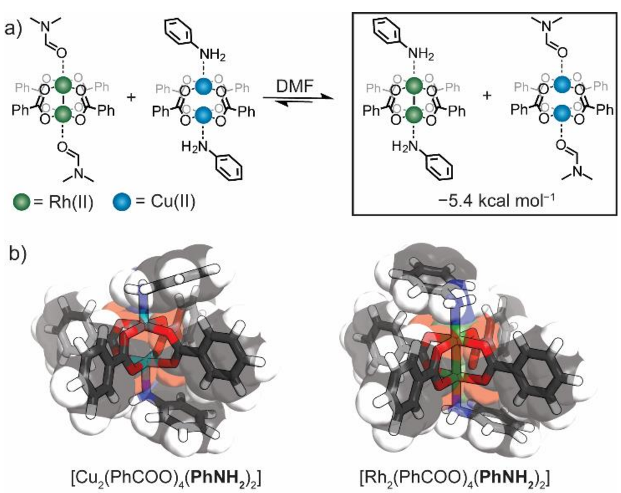

Figure 5: a) The ligand exchange reaction between Cu and Rh paddle-wheel complexes as investigated by DFT calculations; b) a comparison of the optimized structures of $[Cu_2(PhCOO)_4(PhNH_2)_2]$ and $[Rh_2(PhCOO)_4(PhNH_2)_2]$.

CONCLUSIONS

In summary, we have reported the pairwise linking of MOCs as an approach to precisely control the coordination environment and distribution of multiple metal sites and ligands within crystalline porous materials. In doing so, we reported the synthesis and structure of two mixed-cage crystalline solids which can be selectively obtained from the same Cu_4L_4 and Rh_4L_4 MOC mixture through stoichiometric control. In the case of the 2-D MOF (**3**), the benefit of linking **1-Cu** and **2-Rh** into a mixed-cage network extends also to an improved porosity compared to the discrete cage counterparts. It is worth noting that the selective coordination of **1-Cu** to **2-Rh** also addresses the major challenge of synthesizing coordination polymers from kinetically inert Rh^{2+} metal ions.[39] Given the number of accessible Rh^{2+} MOCs,[40,41] we expect that **1-Cu** will serve as a platform for rationally designing multi-functional porous frameworks of higher connectivity and dimentionaility.[42] Indeed, efforts in this direction are currently underway in our laboratory.

ASSOCIATED CONTENT

Supporting Information.

Full details of synthesis, sample preparation, characterization results, crystallographic data and DFT calculations.

Accession codes

CCDC 2106776 – 2106778 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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